Bei Xiao

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Website: claire-bei.github.io | OCTA project on GitHub

RESEARCH PROFILE

Ophthalmology researcher with a professional master's degree in clinical medicine (ophthalmology) and first-author publications in quantitative retinal and choroidal imaging. Experienced in ultra-widefield SS-OCTA, vascular phenotyping, image-quality assessment, and clinical data analysis. Developed Python/OpenCV imaging workflows and a PyTorch/ResNet18 fundus-photograph chronological-age prediction project, including held-out evaluation and a publicly accessible Streamlit demo.

RESEARCH INTERESTS

AI-assisted analysis of multimodal retinal imaging; quantitative imaging biomarkers; retinal and macular disease phenotyping; image-based prediction of disease progression.

EDUCATION

Professional Master's Degree in Clinical Medicine (Ophthalmology)

Southern Medical University, Guangzhou, China | Sep 2021-Jun 2024

Bachelor of Medicine in Clinical Medicine

Hubei University of Science and Technology, Xianning, China | Sep 2016-Jun 2021

RESEARCH EXPERIENCE

Computational Reanalysis of Human CNV Transcriptomic Data

May 2026-present

Independently reanalysed public CNV datasets in R using preprocessing, PCA, differential expression, and functional enrichment. Interpreted immune and extracellular-matrix remodelling signals; manuscript submitted as first and corresponding author.

Quantitative Retinal and Choroidal Imaging in Pachychoroid Spectrum Diseases

Master's research | Sep 2021-Jun 2024

Conducted ultra-widefield SS-OCTA research with image QC, segmentation review, retinal/choroidal measurements, and vascular phenotyping. Characterised pachyvessels, vortex vein dilatation, and venous drainage patterns. Contributed to study design, clinical imaging analyses, and first-author publications.

COMPUTATIONAL PROJECTS

FundusAgeAI: Fundus-Photograph Age Prediction Demo

2026-present | Live online demo

Implemented an ImageNet-pretrained ResNet18 chronological-age regressor using 6,968 ODIR-5K fundus photographs (3,484 patients), patient-level training/validation/test splits, and staged fine-tuning.
Internal held-out evaluation: 523 patients (1,046 images); MAE 8.03 years (95% bootstrap CI 7.54-8.53), versus 8.83 years for a fixed 59-year baseline.
Quality and bias: Assessed age-dependent bias and image-quality effects; manual QC of 200 validation images and within-patient analysis of 36 discordant-quality fellow-eye pairs.
Deployment: Streamlit research demo with the frozen, SHA-256-verified model and original preprocessing. Live Demo: fundusageai-claire-bei.streamlit.app

OCTA Image-Processing Demo

Python / OpenCV | Exploratory project

Developed scripts for OCTA enhancement, binarisation, vessel skeletonisation, and vessel-width estimation. View repository.

ACADEMIC APPOINTMENTS

Guangzhou Xinhua University | Faculty Member | Sep 2026-present
Department of Optometry, School of Medicine and Health; optometry teaching and practical-session coordination.

Guangzhou Huali College | Teaching Faculty | Nov 2025-Jul 2026
Medical-related courses and undergraduate teaching activities.

Bei Xiao

Publications & research skills

PUBLISHED ARTICLES

1. **Xiao B**, Song YP, Yan M, Ye Y, Huang Z. Quantitative Analysis of Choroidal Vascular Structures and Anatomical Changes in Pachychoroid Spectrum Diseases Using Ultra-Widefield SS-OCTA. *Scientific Reports. 2025;15:344.* 10.1038/s41598-024-82745-5

2. **Xiao B**, Yan M, Song YP, Ye Y, Huang Z. Quantitative Assessment of Choroidal Parameters and Retinal Thickness in Central Serous Chorioretinopathy Using Ultra-Widefield Swept-Source Optical Coherence Tomography: A Cross-Sectional Study. *BMC Ophthalmology. 2024;24:176.* 10.1186/s12886-024-03405-w

3. **Xiao B**, Song YP, Ye Y, Huang Z, Yan M. Imaging Characteristics of Affected and Fellow Eyes in Patients with Central Serous Chorioretinopathy at Different Ages. *International Eye Science. 2024;24(6):937-942.* 10.3980/j.issn.1672-5123.2024.6.19

4. **Xiao B**, Song YP, Yan M, Ye Y, Huang Z. A Comparative Study of Choroidal Parameters in the Fellow Eyes of Patients with Pachychoroid Neovasculopathy and Central Serous Chorioretinopathy. *Recent Advances in Ophthalmology. 2024;44(5):376-381.* 10.13389/j.cnki.rao.2024.0073

5. **Xiao B**, Song YP, Yan M, Ye Y, Huang Z. The role of vortex veins in the pathogenesis of central serous chorioretinopathy. *Chinese Journal of Ocular Fundus Diseases. 2024;40(4):319-323.* 10.3760/cma.j.cn511434-20231024-00430

6. **Xiao B**, Song YP, Ye Y, Huang Z, Yan M. Association Between Vortex Vein Dilatation Patterns and Choroidal Thickness Changes in Patients with Central Serous Chorioretinopathy. *Chinese Journal of Ocular Fundus Diseases. 2024;40(5):360-365.* SinoMed article record: 2024322835

7. **Xiao B**, Song YP, Yan M, Ye Y, Huang Z, Chen C, Deng YM. Observation on the Efficacy of Subthreshold Micropulse Laser Photocoagulation in the Treatment of Central Serous Chorioretinopathy Based on Multimodal Imaging Classification. *Recent Advances in Ophthalmology. 2024;44(7):531-535.* 10.13389/j.cnki.rao.2024.0102

MANUSCRIPTS UNDER SUBMISSION

Xiao B, Lu J. AI-Based Retinal Oculomics: A Translational Evidence-Chain Review from Ocular Detection to Systemic Risk Stratification. *Technology and Health Care.* Revised manuscript submitted. First and corresponding author.

Xiao B, Lu J, Zhang X. Computational Reanalysis of Human Choroidal Neovascularization Transcriptomes Identifies Immune-Remodeling Candidate Signals. *Computational Biology and Chemistry.* Submitted. First and corresponding author.

TECHNICAL SKILLS

Imaging: Ultra-widefield SS-OCTA/OCT; image QC; segmentation review; retinal/choroidal measurements.
Programming: Python, OpenCV, PyTorch/ResNet18, Streamlit, NumPy, pandas, Matplotlib, Jupyter; local/cloud inference.
Analysis: Patient-level evaluation and bootstrap; R, SPSS, Stata, GraphPad Prism; regression, PCA, differential expression, functional enrichment.
Laboratory collaborations: Hands-on qPCR, Western blotting, RNA extraction, cell culture, and transfection during master's laboratory collaborations.
Languages: Mandarin Chinese (native); English (IELTS Academic overall 6.5).